

Scalable, High-Capacity Optical Transmission for AI Data Centre Networks

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Abstract This paper reviews ultrawideband optical transmission technologies for scalable AI data centre networks and presents an optical-gateway-based scale-across architecture. Exploiting O-band expansion, WDM/SDM aggregation, and RDMA-over-WDM, the approach enables capacity scaling beyond single-facility limits whilst reducing fibre deployment. ©2026 The Author(s)

Introduction

The rapid evolution of artificial intelligence has driven an unprecedented surge in the deployment of GPU clusters, with demand increasing exponentially as model sizes and training workloads continue to scale [1]. This growth, however, exposes a fundamental constraint of mass data consolidation: the physical and operational limits of individual data centres. Power delivery infrastructure, thermal management capacity, and available floor space impose hard ceilings on how large a single facility can grow. These limitations drive the need for tightly interconnected data centres that can operate as a unified compute fabric.

To overcome these constraints, operators are increasingly adopting scale-across architectures, in which multiple geographically proximate data centres are interconnected to function as a unified system [2–9]. This architectural shift transforms the role of inter-data-centre networks from a supporting utility into a critical performance bottleneck. As GPU clusters are distributed across separated facilities, the network must sustain massive east-west traffic volumes with high reliability and zero packet loss to preserve the efficiency of distributed training and inference. Consequently, the demand for ultrawideband (UWB) wavelength-division multiplexing (WDM), high-capacity transmission technologies in scale-across networks is accelerating rapidly [10–18]. Scale-across systems must deliver not only higher throughput but also improved scalability and operational flexibility to support the next generation of AI-driven workloads, enabling continued growth in computational capability beyond the physical boundaries of individual data centres.

In this paper, we review UWB transmission trials highlighting O-band potentials to scale data centre interconnections. Furthermore, we present a scale-across demonstration using optical gateway switches to address the network requirements of AI workloads, including guaranteed bandwidth, low and predictable latency, and high connectivity whilst reducing the need for fibre deployments.

Spectral scalability

Figure 1 shows the wavelength-dependent attenuation and chromatic dispersion of three cable samples accommodating ITU-T G.657.A1-compliant single-mode fibres [19, 20]. The supported wavelength ranges of various doped-fibre amplifiers are also illustrated by arrows [21, 22]. The attenuation curves follow the combined effects of Rayleigh scattering, OH absorption, and infrared absorption. The lowest attenuation occurs in the C- and L-bands, where transmission loss can be efficiently compensated by erbium-doped fibre amplifiers (EDFAs). These EDFAs are typically designed separately for each band but can be operated in parallel using a band multiplexer.

S-band amplification can be achieved using thulium-doped fibre amplifiers (TDFAs). Because the attenuation over the S-to-L-band region is lower than that of other wavelength regions, these bands are commonly considered for long-distance transmission. In contrast, although the attenuation in the O-band is higher than in the E- and U-bands, chromatic dispersion in the O-band is significantly lower. As a result, the O-band is attractive for power-efficient, high-speed transmission with low-complexity signal processing, particularly in short-reach applications such as intra-data-centre networks [23].

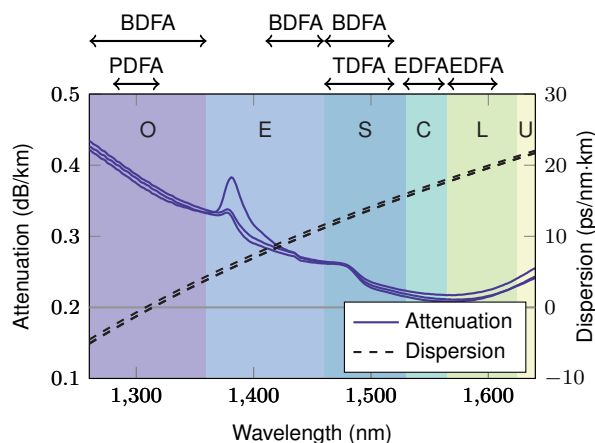


Fig. 1: Sample transmission windows of single-mode fibres.

Praseodymium-doped fibre amplifiers (PDFAs) can compensate for transmission loss in the O-band, particularly near its centre where the chromatic dispersion approaches zero, making this band suitable for intensity modulated direct detection (IM-DD) signals or more advanced 1.6T-class coherent signals that are being considered for standardisation. In addition, bismuth-doped fibre amplifiers (BDFAs) have been reported for broadband amplification from the O- to S-band using various device designs. In particular, O-band BDFAs have been demonstrated with gain bandwidths covering the entire O-band (approximately 17.5 THz), enabling extended-reach 400G-class links based on Coarse WDM (CWDM) or Local Area Network WDM (LAN-WDM) signals [24, 25].

Such O-band signals can co-propagate with signals in the S-, C-, and L-bands, and even with U-band signals [14, 26]. In [27], an O-band signal path partially shared the same fibre with S-, C-, and L-band signals. This approach enables additional O-band paths to be overlaid onto in-service fibre cables without requiring new fibre deployment. Similarly, O-to-U-band transmission systems have been demonstrated by combining multiple types of doped-fibre amplifiers, Raman amplification, and wavelength conversion techniques. Using these approaches, record transmission capacities exceeding 400 Tb/s per fibre have been achieved [15, 18].

These capacity enhancements in UWB systems were realised by improving the output power and gain bandwidth of optical amplifiers. In particular, the pioneering use of the O-band spectrum for coherent WDM transmission broke the capacity limit imposed by high nonlinearity originating from low chromatic dispersion. Due to high nonlinearity, the O-band was not previously considered for high-capacity transmission especially for longer distances. However, a paper firstly demonstrated that 9.6-THz bandwidth equivalent to C+L-band in O-band was utilised for coherent WDM transmission by suppressing the nonlinearity. The nonlinearity was simply mitigated by optimising the launch power. This optimisation was made possible by accurate mathematical modelling accounting for near-zero dispersion and its associated nonlinear effects in the O-band [28–33]. In [16, 18], the entire O-band was exploited for WDM transmission, achieving a usable bandwidth equivalent to three times that of the conventional C-band, or even greater.

Figure 2 plots the generalised mutual information (GMI)-based per-fibre capacities as a function of transmission distance. Multi-band transmission experiments have demonstrated record-high capacities in optical transmission systems. Notably, these records show a significant jump when

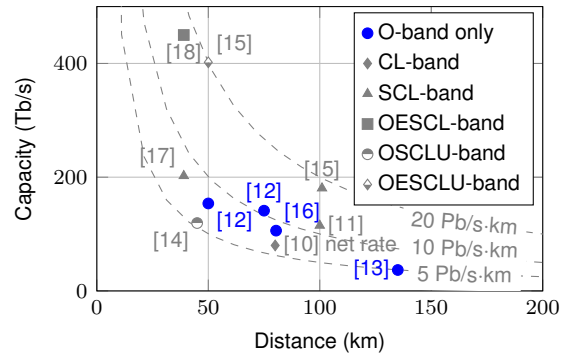


Fig. 2: Per-fibre capacity and distance

O-band multiplexing is employed. This is because the O-band offers a notable increase in usable bandwidth; indeed, O-band transmission experiments have achieved capacity–distance products of around 10 Pb/s-km [12, 13, 16], which are competitive with state-of-the-art SCL-band demonstrations. Further per-fibre capacity increase is possible through space-division multiplexing (SDM). This is also where O-band is useful as the shorter wavelength means there is lower crosstalk one of the main penalties associated with SDM. A total aggregate bandwidth of more than 115 THz using a 12-core fibre specifically designed for operation in the O-band [34].

Scale-across architecture with optical gateway

Large-scale AI training increasingly requires scale-across GPU interconnection beyond a single data centre, yet conventional OTN/ROADM-based metro networks lack transparency, scalability, and latency determinism [2–5, 35]. UWB transmission systems provide a great scalability for data centre network. We propose an optical-circuit-switch (OCS)-based optical gateway architecture enabling Remote Direct Memory Access (RDMA)-over-WDM across geographically distributed GPU clusters whilst reducing required fibre count via WDM/SDM aggregation and supporting metro-scale reach [36].

The proposed architecture adopts a three-tier topology based on a 2D-lattice cabling structure, where clusters are partitioned into overlapping groups with fully connected intra-group physical connectivity. Inter-group connectivity is realized via wavelength-level cross-group switching within OCS gateways. This design ensures that any two clusters are connected with at most one wavelength cross-group switch, achieving a logically full-mesh topology at the WDM layer. The system leverages ZR pluggable transceivers to directly interface scale-up Ethernet/RDMA switches with the optical gateways, enabling RDMA-over-WDM transmission without encapsulation into OTN frames. This eliminates protocol overhead, reduces latency jitter, and avoids congestion caused by sharing resources with legacy services. By

combining WDM and space-division multiplexing (SDM), the architecture significantly reduces fibre deployment requirements from $O(N^2)$ to $O(N)$ whilst maintaining high connectivity. The OCS gateway integrates passive MUX/DEMUX components, optical amplifiers, and an active wavelength routing fabric (implemented using WSS modules in the prototype). This enables dynamic wavelength reconfiguration and flexible capacity allocation without interrupting data-link operation. The architecture supports transmission distances of up to 80 km between adjacent clusters, as defined by Open ZR specifications, and allows metro-scale interconnect diameters exceeding 300 km using SDM interfaces such as MPO-12.

A prototype system interconnecting four GPU clusters was experimentally demonstrated over fibre reaches of 20–30 km per hop. Each cluster employed NVIDIA H100 GPUs connected via RDMA over converged Ethernet (RoCE) switches and 400G ZR modules to the optical gateways. Both RDMA benchmarks and distributed AI training workloads were evaluated. Measured RDMA latency increased linearly with fibre length, as expected from propagation delay; however, by assigning each GPU communication flow to a dedicated wavelength-specific optical path, any network contention was eliminated. Consequently, AI job completion times—evaluated using All-to-All and Bi-directional Ring AllReduce communication patterns—remained essentially unchanged compared to local, single-data-centre deployment.

Conclusions

This paper has reviewed recent progress in ultra-wideband optical transmission for AI data centre networks and highlighted the role of O-band expansion in achieving competitive capacity–distance performance. By combining wavelength-division multiplexing (WDM) and space-division multiplexing (SDM), ultra high per-fibre capacities can be realised whilst mitigating the need for fibre deployment. Furthermore, an optical-circuit-switch (OCS)-based optical gateway architecture was presented to support scale-across GPU interconnection with guaranteed bandwidth, low latency, and high connectivity. Experimental demonstrations show that distributed AI workloads can be executed across metro-scale distances without performance degradation. As available wavelength channels continue to increase, the proposed architecture provides a scalable path to expand capacity and cluster size without additional fibres, supporting the continued growth of AI infrastructure.

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